Network’s Lab.

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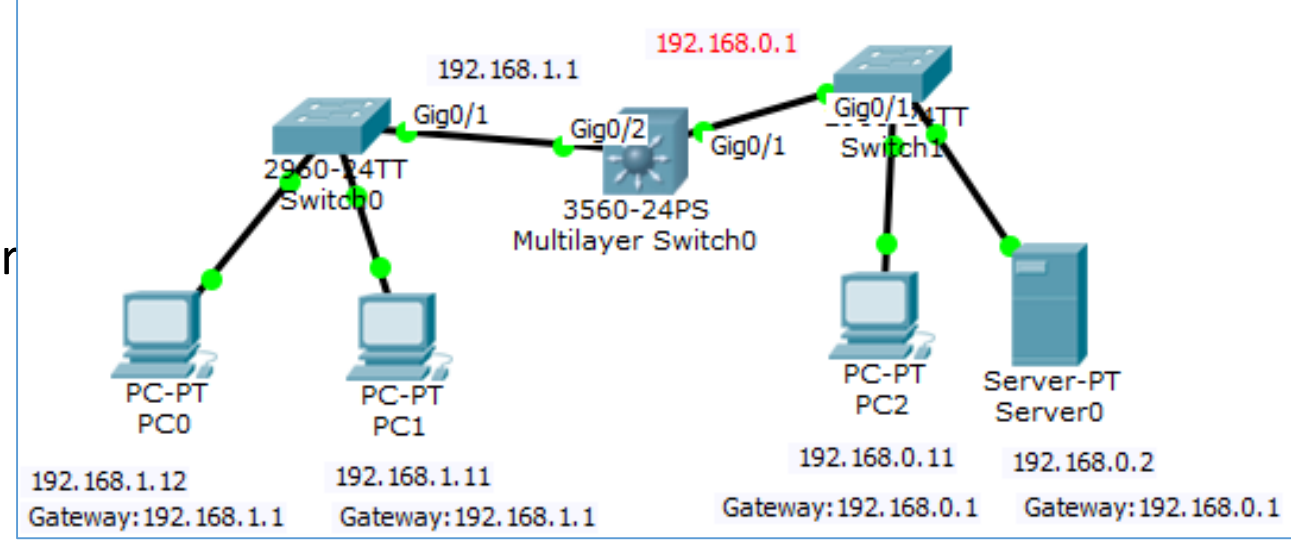
Session: 7

S7_1. One L3 Switch



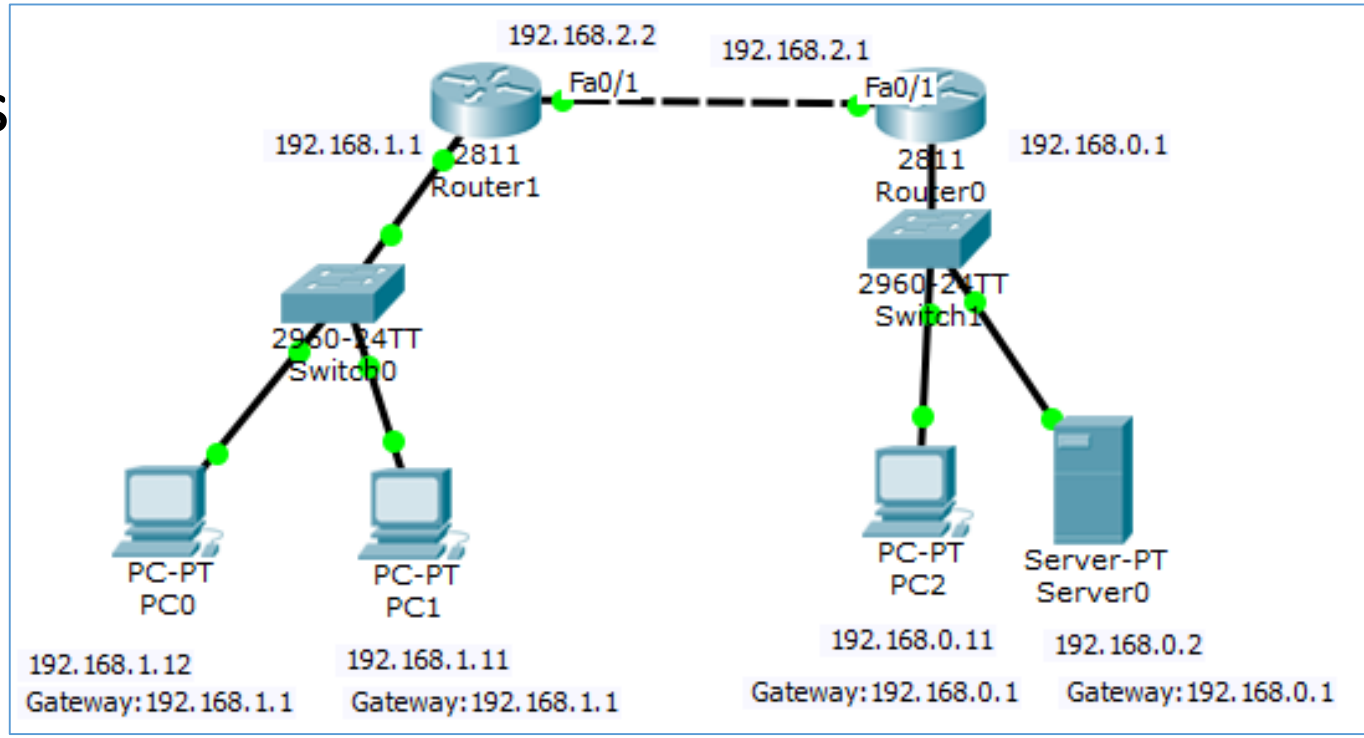
- From S6_3, Change router To L3 Switch
- Try to ping among Networks
- We need to assign IP to each GE port in L3 Sw.
  - L3-Switch#conf t
  - L3-Switch(config)#interface gigabitEthernet 0/1
  - L3-Switch(config-if)#**no switchport**
  - L3-Switch(config-if)#ip address 192.168.0.1 255.255.255.0
  - L3-Switch(config-if)#no shutdown
  - L3-Switch(config-if)#exit
  - ***Set gigabitEthernet 0/2 IP too***
- L3-Switch#show ip interface brief
- Ping From PCs to Gateway address (L3 switch)

S7_1. One L3 Switch

- Ping Between Computers in two different networks
  - It does not work because **Routing is not enables in L3 switches**
- L3-Switch#show ip route
- Enable routing at the L3 switch by
  - L3-Switch(config)#**ip routing**
- L3-Switch#show ip route
- Ping Between Computers in two different networks
  - It works because Two networks are directly connected to the router's ports

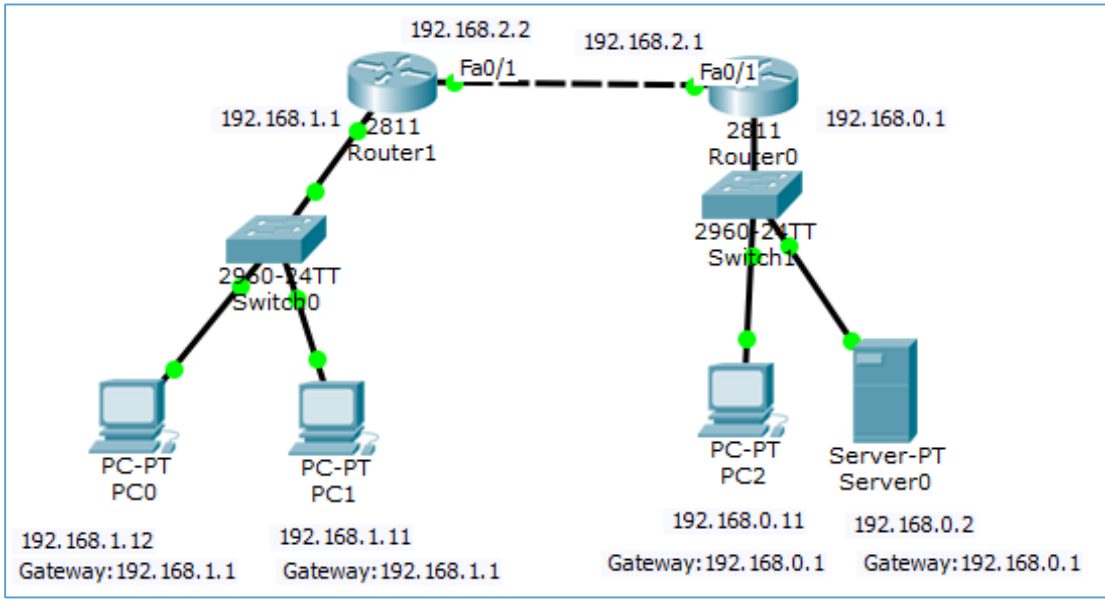
S7_2. Static routes

- Start with S6_3
  add another router



- Set Router's IP addresses as Above, example:
  - Router1(config)#interface fastEthernet 0/1
  - Router1(config-if)#ip address 192.168.2.2 255.255.255.0
  - Router1(config-if)#no shutdown
- From Server, ping followings:
  - Ping 192.168.0.11, Ping 192.168.0.1, Ping 192.168.2.1,
  - Ping 192.168.2.2, Ping 192.168.1.1, Ping 192.168.1.11,

S7_2. Static routes



- We need to add **static routes** to Fix the problem

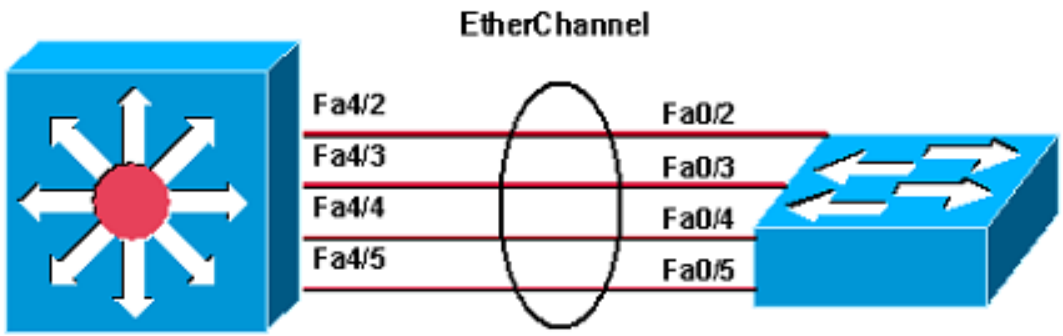
- Adding static route format:
  - Router(config)# ip route <Destination Network Address> <Destination NetworkMask> <Gateway IP/Output Interface>
- Add static Routes:
  - Router0(config)#ip route 192.168.1.0 255.255.255.0 **192.168.2.2**
  - Router0#show ip route
  - Router1(config)#ip route 192.168.0.0 255.255.255.0 **fastEthernet 0/1**
  - Router1#show ip route
- From Server, ping followings:
  - Ping 192.168.0.11, Ping 192.168.0.1, Ping 192.168.2.1,
  - Ping 192.168.2.2, Ping 192.168.1.1, Ping 192.168.1.11,

Do it now!

1. Change **S7_1** as follows:
   - Add another L3_switch
   - Connect two L3 switches and set their interfaces IPs similar to S7_2
   - Add static routes to both L3 switches to work like S7_2

EtherChannel

- **EtherChannel** is a link aggregation technology.
  - It allows grouping of several **physical Ethernet links** to create **one logical Ethernet** link.
- Purpose:
  - providing fault-tolerance
  - high-speed links between switches, routers.
- An EtherChannel can be created from
  - **two to eight** <u>active</u> Fast, Gigabit or 10-Gigabit Ethernet ports,
  - with an additional one to eight <u>inactive</u> (failover) ports
    - which become active as the other active ports fail.
- EtherChannel is primarily used in the backbone network



EtherChannel Management Protocols

- **PAgP** (Port Aggregation Protocol)
  - Is a Cisco protocol
  - is used for the automated, logical aggregation of Ethernet switch ports, known an **etherchannel**.
  - Can only be used between Cisco switches

- **LACP** (Link Aggregation Control Protocol)
  - IEEE specification
  - Provides a method to control the bundling of several physical ports together to form a single logical channel
  - LACP allows a network device to <u>negotiate</u> an automatic bundling of links by sending LACP packets to the peers
  - LACP can have the port-channel load-balance mode

EtherChannel Management Protocols

- **PAgP** Modes:
  - **Auto** : Places an interface into a passive negotiation state, in which the interface responds to **PAgP** packets that the interface receives but does not start PAgP packet negotiation.
    - This setting minimizes the transmission of PAgP packets.
  - **Desirable**: Places an interface into an active negotiation state, in which the interface starts negotiations with other interfaces through the send of **PAgP** packets.
  - **On** : Forces the interface into an EtherChannel **without PAgP** or **LACP**.
    - With the on mode, a usable EtherChannel exists only when an interface group in the on mode has a connection to another interface group in the on mode.
- **LACP** Modes
  - **Active** : Places an interface into an active negotiation state, in which the interface starts negotiations with other interfaces through the send of **LACP** packets.
  - **Passive** : Places an interface into a passive negotiation state, in which the interface responds to **LACP** packets that the interface receives, but does not start LACP packet negotiation.
    - This setting minimizes the transmission of LACP packets.

EtherChannel Management Protocols

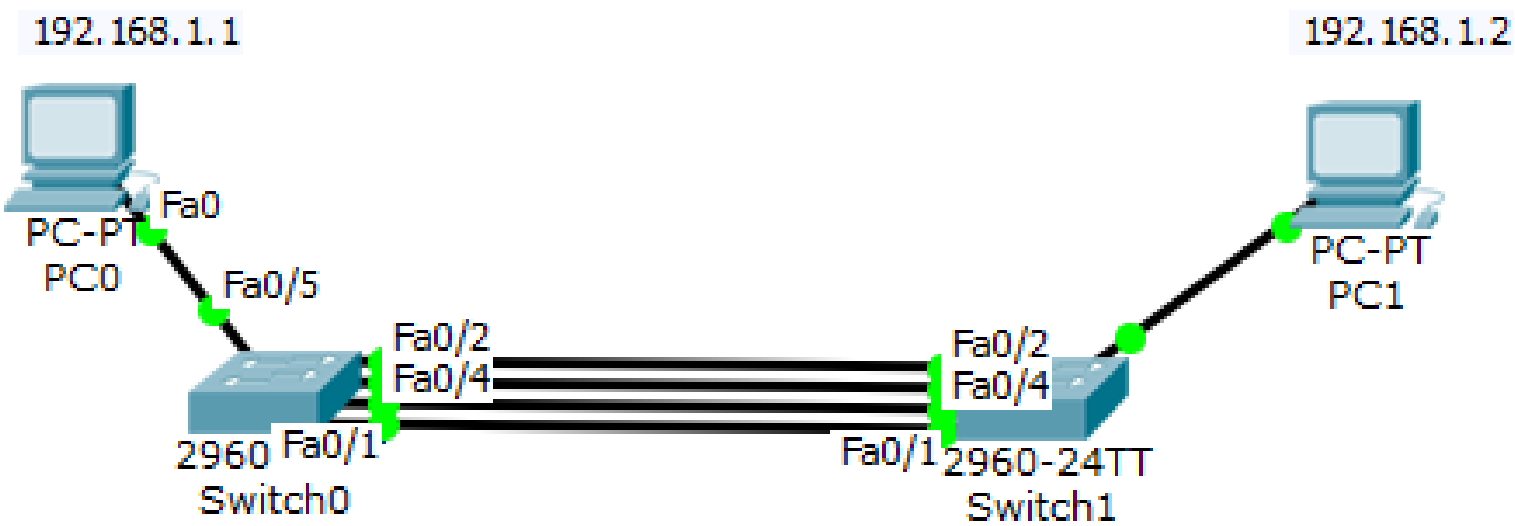
- Switch interfaces exchange **PAgP** packets only with partner interfaces with the auto or desirable mode configuration.
- Switch interfaces exchange **LACP** packets only with partner interfaces with the active or passive mode configuration.
- Interfaces with the **on** mode configuration do **not** exchange PAgP or LACP packets.

There are only three valid combinations to run the LACP link aggregate, as this table shows:

Switch	Switch	Comments
active	active	Recommended.
active	passive	Link aggregation occurs if negotiation is successful.
on	on	Link aggregation occurs without LACP. Although this works, this combination is not recommended.

S7_3.EtherChannel

- Create the following scenario



- Fa0/1 to Fa0/4 are connected to both switches

S7_4.EtherChannel

- On switch0
  - Add port-channel
    - Switch0(config)#interface port-channel 1
      - Note that you can add up to 6 port-channels: 1 to 6
    - Switch0(config-if)#exit
  - Assign Fast ethetnet 0/1 to 0/4 to port-channel 1
    - Switch0(config)#interface range fastEthernet 0/1-4
    - Switch0(config-if-range)#channel-group **1** mode **active**
- Do exactly the above to switch1
- Show commands:
  - Switch0# Show Etherchannel
  - Switch0# Show Etherchannel Summary
  - Switch0# Show Etherchannel Port-Channel

Introduction to Wireshark (With an example log at a wired network)

- Please download and install **Wireshark** from: (available on the ftp)
  - https://www.wireshark.org/
- We will say: How to start, save, load, filter and interpret the log.

*Local Area Connection

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

stp

No.	Time	Source	Destination	Protocol	Length	Info
15	0.341928	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004
111	2.344101	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004
189	4.346275	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004
342	6.348481	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004
433	8.350617	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004
591	10.352785	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004
704	12.354994	Routerbo_2a:ef:a2	Spanning-tree-(for-bridges)_00	STP	60	RST. Root = 32768/0/e4:8d:8c:2a:ef:a5 Cost = 0 Port = 0x8004

```
⊟ Spanning Tree Protocol
    Protocol Identifier: Spanning Tree Protocol (0x0000)
    Protocol Version Identifier: Rapid Spanning Tree (2)
    BPDU Type: Rapid/Multiple Spanning Tree (0x02)
  ⊟ BPDU flags: 0x3c, Forwarding, Learning, Port Role: Designated
      0... .... = Topology Change Acknowledgment: No
      .0.. .... = Agreement: No
      ..1. .... = Forwarding: Yes
      ...1 .... = Learning: Yes
      .... 11.. = Port Role: Designated (3)
      .... ..0. = Proposal: No
      .... ...0 = Topology Change: No
  ⊟ Root Identifier: 32768 / 0 / e4:8d:8c:2a:ef:a5
      Root Bridge Priority: 32768
      Root Bridge System ID Extension: 0
      Root Bridge System ID: Routerbo_2a:ef:a5 (e4:8d:8c:2a:ef:a5)
    Root Path Cost: 0
  ⊟ Bridge Identifier: 32768 / 0 / e4:8d:8c:2a:ef:a5
      Bridge Priority: 32768
      Bridge System ID Extension: 0
      Bridge System ID: Routerbo_2a:ef:a5 (e4:8d:8c:2a:ef:a5)
```

Home Work

Using Wireshark :

- Save a log from a wired network and filter and find STP protocol's packets
- **OR**
- Save a log when you are pinging an online server. Now show (filter) ICMP packets that are sent to the server and returned from the server

Bonus assignment (optional): Using Wireshark:

- Save a log when you are submitting a login form (must be http, not https) . Now Find and show http request packet's header which show your login form detail: (username and password)